Noctua: Unsupervised Network Intrusion Detection under Training Contamination, Temporal Drift and Dataset Shift

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***Abstract*—Unsupervised detectors are attractive for network intrusion detection because attack labels are scarce and attacks evolve, yet they are routinely evaluated on a clean training window, stationary traffic and a single network. We study what remains of their accuracy, and of their false-alarm guarantees, when these three assumptions are removed one at a time. Fifteen detectors spanning tree, kernel, density, distance, reconstruction, deep one-class and self-supervised families are fitted without labels on NSL-KDD, UNSW-NB15 and CIC-IDS2017 and calibrated with split-conformal p-values, so that every score carries a finite-sample false-positive guarantee that can be put to the test. Hidden attacks making up only 5% of the training window cost 11.4 ROC-AUC points on average (up to 36). We propose Consensus-Trimmed Refit (CTR), a label-free screen by three independent detectors before fitting, which recovers +5.4 points on average and improves all 21 dataset–detector pairs at a cost of 2.5 points on clean data. Conformal calibration holds only approximately even in-dataset (single detectors realize 2.1–4.9% false positives at a nominal 1%), and it degrades under drift (1.6% to 7.6% across days of CIC-IDS2017) and shift (cross-dataset AUC of 0.695 and 0.446). The population stability index of the benign score distribution tracks the loss of the guarantee (Spearman $r_s = 0.52$), and a diverse five-detector conformal merge stays within budget at 41–67% recall. We release the protocol, per-seed results, code and an interactive laboratory.**

***Index Terms*—Network intrusion detection, unsupervised anomaly detection, conformal prediction, training-data contamination, concept drift, dataset shift, benchmark.**

I. Introduction

ANOMALY detection is the natural formulation of network intrusion detection when attacks are rare, heterogeneous and constantly changing: normal behaviour is abundant and can be modelled, while labelled attacks are scarce, expensive and quickly outdated [1]–[3]. Unsupervised detectors, from isolation forests to deep autoencoders and normalizing flows, therefore dominate the research literature, and published results on corpora such as NSL-KDD, UNSW-NB15 and CIC-IDS2017 often report very high ROC-AUC.

Such numbers are obtained under conditions that a deployment cannot guarantee. The training window is assumed to be *clean* (attack-free), although the very reason for using an unsupervised model is that nobody can label the traffic. Test traffic is assumed to follow the *training distribution*, although networks change from one day to the next. And a single *dataset* is used to claim generality. When these assumptions fail, a headline AUC says little about what an operator experiences: the *false-alarm rate* at a threshold that was fixed in advance.

We therefore ask three questions about unsupervised network detectors.

- **Q1 Contamination.** How much does a small fraction of undetected attacks in the training window hurt each detector family, and can the damage be repaired without labels?
- **Q2 Drift.** How quickly does a threshold calibrated on one day stop meeting its false-positive budget, and is the breakdown visible without labels?
- **Q3 Shift.** What survives when a detector trained on one network is deployed on another?

To make answers to Q2 and Q3 interpretable we attach a statistical guarantee to every detector. *Split-conformal* calibration turns any anomaly score into a p-value whose false-positive rate is bounded in finite samples under exchangeability [4], [5]. The guarantee is exact when exchangeability holds, so its realized violation is a direct, unit-free measure of how far a deployment has moved from the calibration data.

Summary of findings

(Q1) Hidden attacks are learned as normal: with 5% contamination the average AUC loss is 11.4 points, and reconstruction, density and distance models are hit hardest (up to 36 points for kNN distance on CIC-IDS2017). A label-free committee screen, CTR, recovers roughly half of the average loss. *(Q2)* Drift is not a smooth decay: the realized false-positive rate of a Monday-calibrated 1% threshold is 1.6% on Wednesday but 7.6% on Thursday, and a population-stability index of the benign scores follows it. *(Q3)* Transfer between UNSW-NB15 and CIC-IDS2017 is poor, and even below chance in one direction.

Contributions

1) A reproducible protocol (four experiments, 15 label-free detectors, fixed configurations, a held-out calibration slice, bootstrap intervals and rank-based tests) over three public datasets that require no registration (Section V).
2) A conformal layer that gives every detector a finite-sample false-positive bound, a valid merge of heterogeneous detectors, and an empirical account of where the bound breaks in-dataset, under drift and under shift (Sections III and VI).

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Code, per-seed results and an interactive laboratory: https://github.com/fragompul/unsupervised-anomaly-network-intrusion.

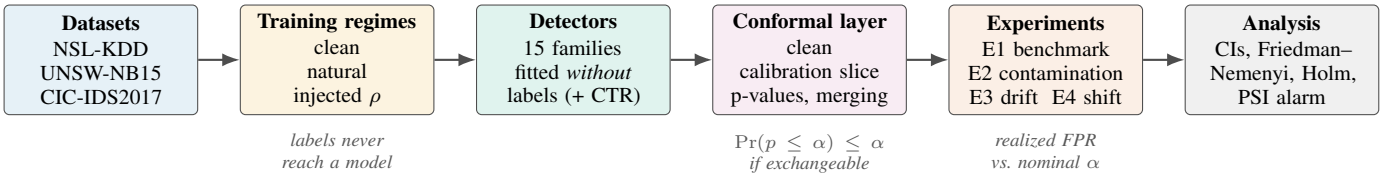


Fig. 1: Overview of the protocol. Labels are used only to score results and to decide which rows enter each training regime, never as model input.

3) *Consensus-Trimmed Refit*, a simple label-free defence against training contamination, with ablations over trim level, screener committee and an oracle trim (Sections IV and VI).
4) A population-stability alarm validated against the breakdown of the conformal guarantee on real timestamps, with its precision and recall at several thresholds (Section VI).
5) Exact reconstruction attributions with a deletion-faithfulness test, and an open interactive laboratory with per-seed results and code.

II. Related Work

Evaluation of learning-based intrusion detection: Sommer and Paxson [1] argued that anomaly detection in networks differs from other machine-learning tasks because of the cost of errors, the diversity of normal traffic and the difficulty of evaluation, and Arp *et al.* [3] catalogued recurring pitfalls such as sampling bias and spurious correlations. TESSERACT [6] showed that ignoring time inflates malware-classification results. We bring that discipline to unsupervised network detectors, with a temporal split for CIC-IDS2017 and an explicit calibration of the alarm threshold. Our datasets are NSL-KDD [7], UNSW-NB15 [8] and CIC-IDS2017 [9], whose labelling and flow-construction problems are documented [10]. Autoencoder ensembles have been proposed for online intrusion detection [11], but under clean-training assumptions.

Anomaly-detection benchmarks and detectors: Surveys [12]–[14] and ADBench [15] compare detectors on tabular data, usually under clean training and i.i.d. splits. Our detector set covers Isolation Forest [16], one-class SVM [17], LOF [18], ECOD [19], COPOD [20] and HBOS [21] (via PyOD [22]), Deep SVDD [23], an autoencoder, a β-VAE [24], [25], a RealNVP flow [26], NeuTraL AD [27] and a masked feature-reconstruction transformer in the spirit of FT-Transformer [28].

Robustness to contaminated training data: Robust PCA [29] and robust deep autoencoders [30] separate a low-rank or low-error component from sparse corruption inside a single model. CTR is complementary: it screens the training set with a committee of cheap, structurally different detectors *before* any model is fitted, so it can wrap any detector in the zoo.

Conformal prediction and drift: Conformal prediction yields distribution-free guarantees under exchangeability [4], [31], [32]; conformal p-values for outlier testing are analysed in [5], and p-values can be merged by averaging [33]. Extensions handle covariate shift [34], non-exchangeable data [35] and online distribution shift [36]. Detecting drift from model outputs is studied in [37], and the broader drift literature is surveyed in [38], [39]. We do not propose a new conformal method; we measure, for fifteen detectors on real network data, how much the basic guarantee degrades and whether a simple label-free statistic anticipates it.

III. Problem Setting

Detectors and the label-free contract: A flow is a vector $x \in \mathbb{R}^d$ with a hidden label $y \in \{0, 1\}$ (1 = attack). A detector is a procedure that, given an unlabeled training matrix X_{fit}, returns a score function $s : \mathbb{R}^d \to \mathbb{R}$, higher meaning more anomalous. Labels are never available to fit; they only score results and decide which rows form each training regime.

Training regimes: Let P_0 be the distribution of normal flows and P_1 that of attacks. A *clean* regime fits on samples of P_0 only; the *natural* regime fits on the training split exactly as released; the *injected* regime fits on $(1-\rho)P_0 + \rho P_1$ with an exact contamination rate ρ.

Split-conformal p-values: A calibration set $\mathcal{C} = \{c_1, \ldots, c_m\}$ of clean normal flows is held out before anything else and never used for fitting. The conformal p-value of a flow x is

$$p(x) = \frac{1 + |\{c \in \mathcal{C} : s(c) \geq s(x)\}|}{m+1}. \tag{1}$$

If a normal test flow is exchangeable with $\mathcal{C}$, then $\Pr(p(X) \leq \alpha) \leq \alpha$ for every $\alpha \in (0, 1)$ and every detector [4]. Flagging a flow when $p(x) \leq \alpha$ therefore bounds the false-positive rate by α *without any assumption on the score distribution*. Let Q denote the distribution of normal flows at deployment and q_α the α-upper quantile of the calibration scores. The realized false-positive rate is

$$\text{FPR}(\alpha) = Q\big(s(X) \geq q_\alpha\big), \tag{2}$$

which equals α (up to the finite-sample slack of Eq. 1) only if $Q = P_0$ in the score space. Contamination, drift and transfer are three ways of making $Q \neq P_0$; the gap $\text{FPR}(\alpha) - \alpha$ measures how much.

Merging detectors: For K detectors with p-values $p_1, \ldots, p_K$ (valid, possibly dependent), $\min\big(1, 2\bar{p}\big)$ with $\bar{p} = K^{-1}\sum_k p_k$ is again a valid p-value [33]. For *ranking* (AUC) we instead use $K^{-1}\sum_k -\log_{10} p_k$, which is not a p-value but does not saturate at the p-value floor.

Population stability index: For reference scores r (the calibration scores) and current scores u, let $B = 10$ bins be defined by the deciles of r, and let r_b, u_b be the fractions of each sample in bin b (clipped away from zero). Then

$$\text{PSI}(r, u) = \sum_{b=1}^{B} (u_b - r_b) \ln \frac{u_b}{r_b} \tag{3}$$

is the population stability index (PSI) [40], conventionally read as moderate shift above 0.1 and major shift above 0.25. Eq. 2 depends on the upper tail of the shifted score distribution, whereas Eq. 3 integrates the discrepancy over all deciles; we test empirically whether the second anticipates the first.

IV. Consensus-Trimmed Refit

Attacks present in the training window are learned as normal by density, reconstruction and distance models, which lowers the scores that those very attacks receive at test time. CTR removes the likely culprits *before* fitting (Algorithm 1). A committee of three structurally different, cheap screeners (Isolation Forest, ECOD and PCA reconstruction) scores the training rows; each score is converted to a percentile rank, the ranks are averaged, and the top τ fraction by consensus rank is dropped. The base detector, which can be any detector of the zoo, is then fitted on the remainder.

Why a committee: Each screener fails differently: PCA reconstruction misses attacks that lie in the principal subspace, ECOD looks at marginal tails only, and Isolation Forest isolates points by axis-aligned splits. Averaging percentile ranks keeps a flow only if *no coalition of inductive biases* finds it extreme, which is more specific than the tail of any single detector. We test this claim by ablation rather than assume it (Section VI).

Residual contamination: Suppose a fraction ρ of the rows are attacks and the trimmed set contains a fraction r_τ of them. By counting, the residual contamination of the retained set is

$$\rho' = \frac{\rho\,(1 - r_\tau)}{1 - \tau}, \tag{4}$$

so trimming lowers contamination ($\rho' < \rho$) if and only if the committee ranks attacks above benign flows better than chance ($r_\tau > \tau$), and it always discards a fraction $\tau - \rho\, r_\tau$ of the rows that are legitimate. The trim τ is thus an assumed upper bound on contamination, and CTR carries a price on clean data that we quantify rather than hide.

Algorithm 1 Consensus-Trimmed Refit

Input: unlabeled training matrix $X \in \mathbb{R}^{n \times d}$, base detector $\mathcal{D}$, screeners $\mathcal{S} = \{S_1, \ldots, S_M\}$, trim $\tau \in [0, 1)$
Output: fitted detector

for $j = 1, \ldots, M$ **do**
  fit S_j on X and compute scores $a_i^{(j)} = S_j(x_i)$
  $\pi_i^{(j)} \leftarrow \mathrm{rank}(a_i^{(j)})/n$ ▷ percentile rank in $(0, 1]$
$g_i \leftarrow M^{-1} \sum_j \pi_i^{(j)}$ ▷ consensus rank
$\theta \leftarrow$ the $(1 - \tau)$-quantile of $\{g_i\}$
$X' \leftarrow \{x_i : g_i \le \theta\}$
return $\mathcal{D}$ fitted on X'

V. Experimental Protocol

A. Datasets

Table I summarizes the corpora. NSL-KDD and UNSW-NB15 use their official train/test partitions. CIC-IDS2017 is split *temporally*: Monday–Tuesday for training (Monday is attack-free, Tuesday adds FTP and SSH brute force at about 1.4% contamination) and Wednesday–Friday for testing, so almost every test attack family is novel to the training window. It is subsampled class-aware (rare classes kept whole), which tilts rare-attack prevalence upward. Identifiers (IP addresses, ports, flow id, timestamps) are never used as features. For cross-dataset experiments UNSW-NB15 and CIC-IDS2017 are mapped to 18 physically aligned flow features (durations, per-direction packets and bytes, rates, mean packet sizes, inter-arrival times and protocol indicators). All downloads are pinned by size and SHA-256 from public mirrors, and require no registration.

B. Detectors

Table II lists the 15 detectors. All implement one interface and use one *fixed* configuration across datasets, with no hyper-parameter chosen on labels: tuning on labels would turn "unsupervised" into a claim about the pipeline rather than the model. Most neural models are trained with AdamW [41] and early stopping on held-out *training* loss under a wall-clock cap, using PyTorch [42]; classical models use scikit-learn [43] and PyOD [22].

C. Calibration and experiments

In every experiment, 15% of the clean normal flows of the training split (at most 20 000) are held out as $\mathcal{C}$ before fitting (E3 uses 15% of Monday; E4 caps $\mathcal{C}$ at 15 000). Assuming a small curated calibration sample is far weaker than assuming a curated training set, and we state it as an assumption. Four experiments remove one assumption each; each cell fits all detectors on one training set and stores *raw score vectors*, from which every number is recomputed.

- **E1, main benchmark.** All detectors on three datasets under the *clean* (normal-only) and *natural* regimes, 5 seeds.
- **E2, contamination.** Attacks injected at rates $\rho \in \{0, 1, 2, 5, 10, 20\}\%$ into a 30 000-row normal base, for ten base detectors and CTR variants. CIC-IDS2017 uses

TABLE I: Datasets. "Enc. feat." is the dimension after encoding and dropping constant columns; "Atk. test/train" is the percentage of attacks in the evaluation and in the released training split.

Dataset	Year	Train	Test	Raw feat.	Enc. feat.	Atk. test (%)	Atk. train (%)
NSL-KDD	2009	125,973	22,544	41	122	56.9	47
UNSW-NB15	2015	175,341	82,332	34	50	55.1	68
CIC-IDS2017	2017	≈150k	123,731	78	69	31.4	≈1.4

TABLE II: Detector zoo. Fit and scoring cost are means over the clean regime of E1 on a laptop CPU (relative, not a deployment benchmark).

Detector	Family	Fixed configuration	Ref.	Fit (s)	Score (ms/1k)
Autoencoder	Reconstruction	MLP 128-64-$\lfloor d/8 \rfloor$, MSE	–	10.1	0.6
Gaussian mixture	Density	8 full-covariance components	–	2.8	1.6
kNN distance	Distance	mean distance to k=10 neighbours	–	0.0	93.4
NeuTraL AD	Self-supervised	K=6 learned transformations	[27]	45.7	7.2
β-VAE	Reconstruction	β=0.5, negative ELBO	[24], [25]	10.1	0.5
Deep SVDD	Deep one-class	bias-free MLP, 10+30 epochs	[23]	6.0	0.1
RealNVP flow	Density	RealNVP, 6 couplings, 128 hidden	[26]	53.5	6.7
Masked Tab-Transformer	Self-supervised	2-layer, 4-head, 30% masking	[28]	155.8	985.8
PCA reconstruction	Reconstruction	95% variance retained	–	0.1	0.3
Local Outlier Factor	Distance	k=20, novelty mode	[18]	2.2	97.4
Isolation Forest	Tree	200 trees, 256 samples/tree	[16]	0.2	7.8
HBOS	Density	20 histogram bins	[21]	0.9	3.2
ECOD	Density	parameter-free ECDF tails	[19]	0.9	52.0
One-Class SVM	Kernel	RBF, ν=0.05, ≤8k rows	[17]	0.2	36.3
COPOD	Density	parameter-free copula tails	[20]	0.7	48.8

its i.i.d. split here so every attack family is available for injection.

- **E3, drift.** Fit and calibrate on Monday of CIC-IDS2017; evaluate Tuesday to Friday with the Monday-calibrated threshold.
- **E4, shift.** Fit on one dataset's harmonized clean normals and detect on the other, with the source scaler or with per-domain standardization computed from the *unlabeled* target.

Seeds vary detector initialization, the calibration split and the contamination draw; dataset splits are fixed.

D. Metrics and statistics

We report ROC-AUC, PR-AUC and TPR at 1% FPR as threshold-free quality, and at the conformal operating point $\alpha \in \{1\%, 5\%\}$ the realized FPR, TPR and F1. The *calibration honesty check* is the realized FPR against the nominal α (Eq. 2). Confidence intervals are bootstrap intervals over seeds. Detectors are ranked with the Friedman test, Nemenyi critical differences [44] and Holm-corrected [45] Wilcoxon tests against the best-ranked method, over (dataset, regime, seed) blocks. Because dataset splits are fixed, seeds are replicates over models and not over datasets, which makes these tests optimistic [46]; we therefore also report the coarser (dataset, regime) analysis.

VI. Results

A. E1: main benchmark and calibration

Table III reports ROC-AUC and Fig. 2 the critical-difference diagram over 30 blocks (Friedman $\chi^2 = 85.6$, CD $= 3.92$). The Autoencoder has the best mean rank (5.0), and it is statistically tied with the next 9 detectors, so a broad set of reconstruction, density, distance and self-supervised detectors is statistically indistinguishable and the evidence does not support a single winner. The best clean-regime AUC is 0.966 (Autoencoder). At the coarser level of (dataset, regime) blocks the Friedman test is not significant ($\chi^2 = 17.9$, $p = 0.21$ over 6 blocks, CD $= 8.76$), which a careful reader should take as the honest bound on how finely one can rank detectors across datasets. The natural regime is harsh: the released UNSW-NB15 training split is 68% attacks, and 14 of 15 single detectors fall below AUC 0.5 there, i.e. they score attacks as *more* normal than normal flows.

Calibration is the more telling comparison. At a nominal $\alpha = 1\%$, the realized false-positive rate of single detectors in the clean regime is 3.0% on NSL-KDD, 2.1% on UNSW-NB15 and 4.9% on CIC-IDS2017 (Fig. 3): even in-dataset, the test traffic is shifted relative to the calibration slice (novel services, later days, unseen attack-adjacent behaviour). Merging detectors with averaged conformal p-values is valid, but its power depends on its members. The diverse five-detector merge (Isolation Forest, kNN, autoencoder, flow, NeuTraL AD) stays within budget (0.2–1.7%) with 41–67% recall at $\alpha = 5\%$, whereas merging all fifteen, weak members such as ECOD and COPOD on UNSW-NB15 included, is so conservative that recall at $\alpha = 5\%$ falls to 1.4% on that dataset.

B. E2: training contamination (Q1)

Fig. 4 and Table IV show the effect of hidden attacks. At $\rho = 5\%$ the mean AUC drop over detector–dataset pairs is 11.4 points, and kNN suffers most, while the autoencoder and the mixture model lose more than ten points on UNSW-NB15 and CIC-IDS2017; all of them estimate their notion of normal

TABLE III: ROC-AUC (mean over seeds) for single detectors and conformal ensembles. C and N denote the clean and natural regimes. Bold: best single detector per column; underlined: second best. "Rank" is the mean rank over all blocks.

	Clean training			Natural training			
Detector	NSL	UNSW	CIC	NSL	UNSW	CIC	Rank
Autoencoder	**0.966**	0.913	0.864	0.804	0.315	0.863	5.0
Gaussian mixture	0.959	0.906	<u>0.873</u>	<u>0.900</u>	0.259	<u>0.877</u>	5.1
kNN distance	<u>0.962</u>	<u>0.916</u>	0.844	0.646	<u>0.439</u>	0.843	6.6
NeuTraL AD	0.951	**0.931**	0.831	0.764	0.439	0.844	7.3
β-VAE	0.962	0.863	**0.880**	0.809	0.251	**0.877**	6.1
Deep SVDD	0.957	0.889	0.855	0.730	0.402	0.849	6.9
RealNVP flow	0.933	0.903	0.853	0.823	0.285	0.854	8.4
Masked Tab-Transformer	0.930	0.859	0.861	0.726	0.369	0.856	8.5
PCA reconstruction	0.955	0.798	0.861	0.815	0.417	0.861	6.6
Local Outlier Factor	0.865	0.825	0.870	0.642	**0.529**	0.867	8.0
Isolation Forest	0.949	0.781	0.811	0.870	0.282	0.808	9.8
HBOS	0.945	0.674	0.724	0.853	0.293	0.722	11.1
ECOD	0.952	0.588	0.731	**0.903**	0.329	0.734	9.4
One-Class SVM	0.886	0.590	0.789	0.840	0.439	0.794	10.0
COPOD	0.949	0.550	0.719	0.897	0.290	0.722	11.3
Ensemble (all 15)	0.961	0.897	0.877	0.872	0.328	0.878	–
Ensemble (diverse 5)	0.960	0.920	0.852	0.821	0.327	0.857	–

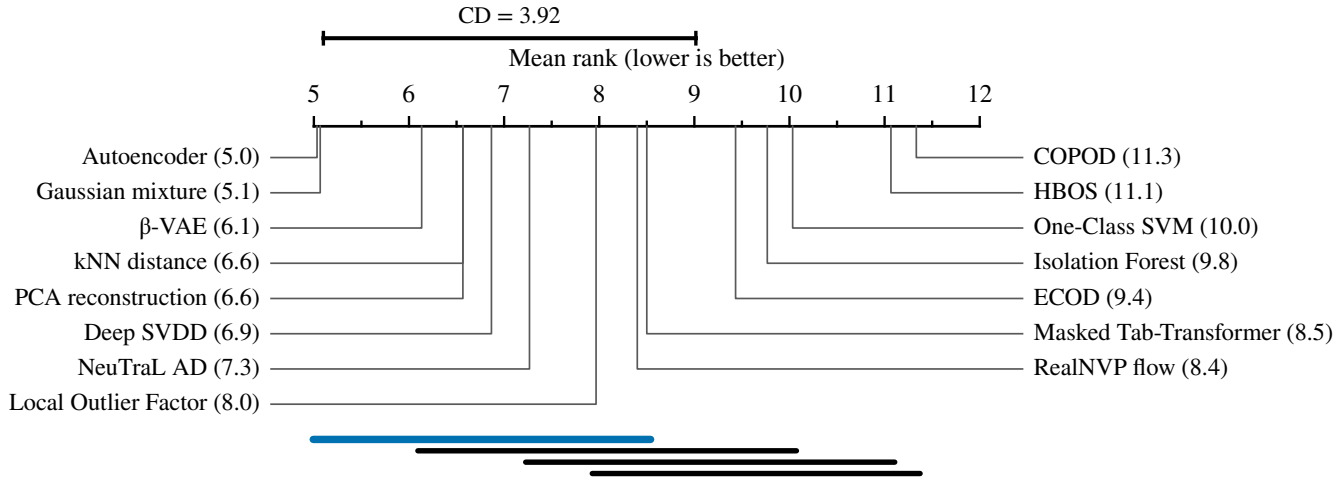


Fig. 2: Critical-difference diagram of ROC-AUC mean ranks over 30 blocks. Detectors joined by a bar are not significantly different (Nemenyi, $\alpha = 0.05$). The blue bar contains the best-ranked detector.

directly from the contaminated training sample. CTR with a 10% trim changes the AUC by +5.4 points on average and improves 21/21 dataset–detector pairs. The gain is +2.3 at $\rho = 1\%$ (16/21), +4.8 at 2% (21/21), +2.2 at 10% (16/21) and +1.4 at 20% (13/21): it fades as contamination exceeds the assumed bound τ, in line with Eq. 4. The price on clean data ($\rho = 0$) is -2.5 points (improving only 2/21 pairs), because rare but legitimate flows are trimmed.

The ablation (Table V) is deliberately reported as found. Committee purification improves the autoencoder at $\rho = 5\%$ on NSL-KDD and CIC-IDS2017 for every trim, whereas on UNSW-NB15 a 5% trim is slightly worse than no purification and larger trims are needed; on UNSW-NB15 and CIC-IDS2017 larger trims help more, while on NSL-KDD the three trims are close. A single well-matched screener can beat the 10% committee (ECOD or Isolation Forest alone on NSL-KDD and UNSW-NB15), while a poorly matched one hurts (PCA alone is worse than the committee on every dataset and

TABLE IV: Effect of 5% contamination and CTR (ROC-AUC, mean over seeds).

Dataset	Detector	ρ=0	ρ=5%	+ CTR	Δ
NSL-KDD	Autoencoder	0.968	0.907	0.944	+0.037
NSL-KDD	kNN distance	0.962	0.812	0.912	+0.100
NSL-KDD	RealNVP flow	0.933	0.918	0.953	+0.034
NSL-KDD	Gaussian mixture	0.964	0.924	0.953	+0.028
UNSW-NB15	Autoencoder	0.911	0.766	0.781	+0.016
UNSW-NB15	kNN distance	0.916	0.634	0.725	+0.091
UNSW-NB15	RealNVP flow	0.902	0.680	0.768	+0.088
UNSW-NB15	Gaussian mixture	0.901	0.651	0.719	+0.068
CIC-IDS2017	Autoencoder	0.887	0.660	0.688	+0.029
CIC-IDS2017	kNN distance	0.807	0.447	0.523	+0.075
CIC-IDS2017	RealNVP flow	0.857	0.705	0.733	+0.028
CIC-IDS2017	Gaussian mixture	0.890	0.754	0.805	+0.051

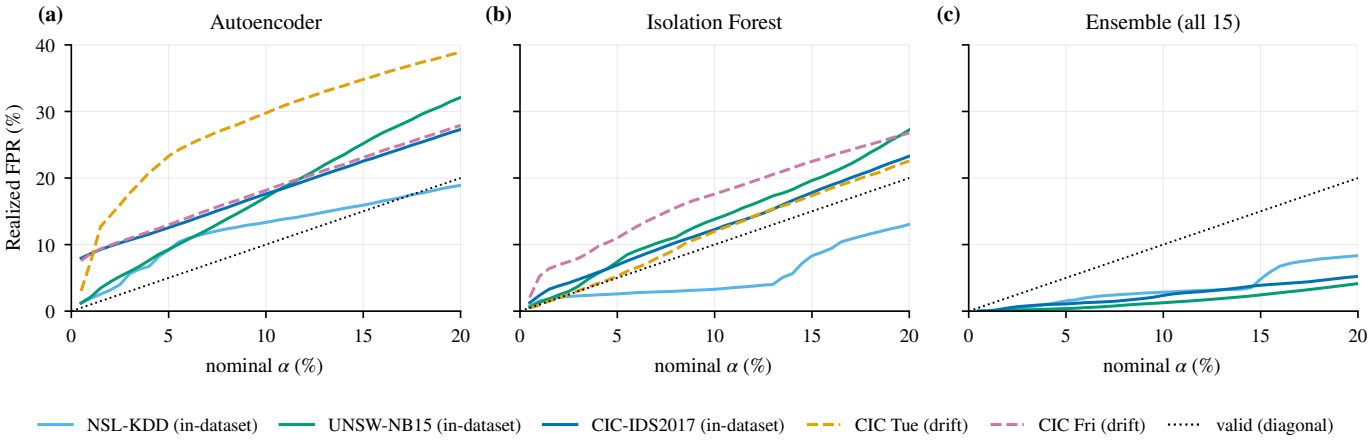


Fig. 3: Reliability diagrams: realized false-positive rate against nominal α for three scorers. Solid: in-dataset evaluation on each corpus; dashed: CIC-IDS2017 days with the Monday-calibrated threshold. The dotted diagonal is the conformal guarantee; above it the promise is broken.

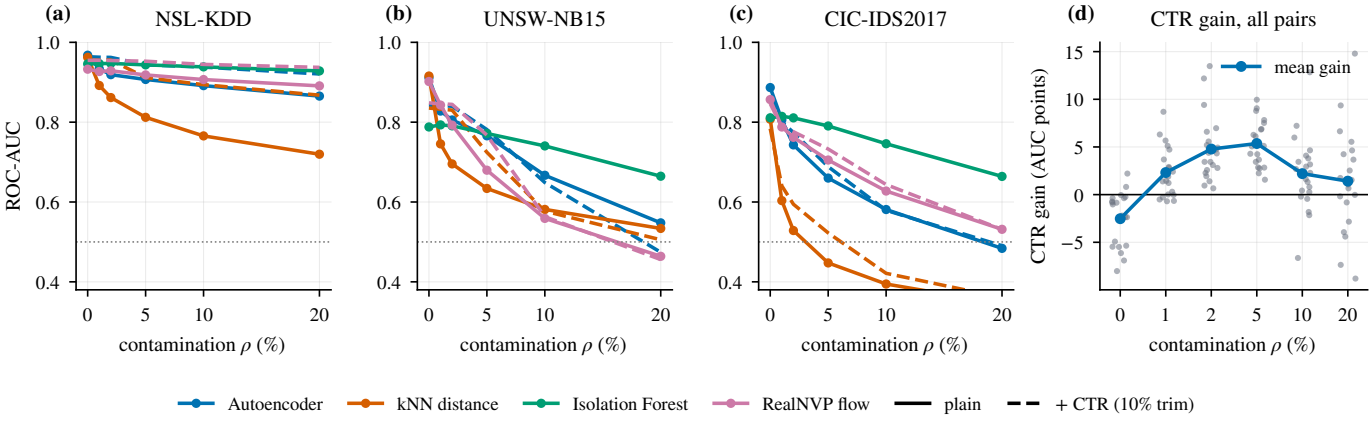


Fig. 4: (a–c) ROC-AUC against training contamination for four detectors; solid lines are plain detectors and dashed lines are the same detectors behind CTR with a 10% trim. (d) CTR gain for every dataset–detector pair (grey points) and their mean (blue).

TABLE V: Ablation of CTR on the autoencoder at $\rho = 5\%$ (ROC-AUC).

Training-set purification	NSL-KDD	UNSW-NB15	CIC-IDS2017
No purification	0.907	0.766	0.660
Committee, 5%	0.947	0.737	0.662
Committee, 10% (default)	0.944	0.781	0.688
Committee, 20%	0.960	0.817	0.697
IForest only, 10%	0.964	0.807	0.683
ECOD only, 10%	0.961	0.827	0.684
PCA only, 10%	0.933	0.678	0.646
Oracle trim (5%)	0.947	0.737	0.662

below no purification on UNSW-NB15 and CIC-IDS2017). The committee therefore buys *robustness to the choice of screener*, not uniform dominance. An oracle trim equal to the true ρ coincides with the 5% trim at this contamination rate.

C. E3: temporal drift (Q2)

Fig. 5(a) shows the realized benign false-positive rate by day for a Monday-calibrated 1% threshold. Drift is *not* monotone in time: averaged over single detectors the rate is 4.7% on Tuesday, 1.6% on Wednesday, 7.6% on Thursday and 6.2% on Friday, because each day introduces its own benign services. Different detectors break on different days. ECOD, COPOD and HBOS stay closest to the budget, whereas RealNVP flow and Deep SVDD break hardest.

The PSI of the benign scores (Eq. 3) correlates with the realized rate (Spearman $r_s = 0.52$, Fig. 5(b)). Used as an alarm for a broken guarantee (realized FPR at least $3\times$ nominal; 144 such detector–day–seed cells), PSI ≥ 0.10 attains recall 58% and precision 76%; PSI ≥ 0.05 attains recall 94% at precision 75%; and the conventional PSI ≥ 0.25 has precision 79% but recall only 13%. The monitor is thus best read as a tunable early warning, and the conventional "major shift" threshold is too conservative for score distributions of this kind. An important caveat: PSI is computed here on the benign flows of each day to isolate covariate drift from attack prevalence; a deployed monitor would see mixed traffic, which we do not evaluate.

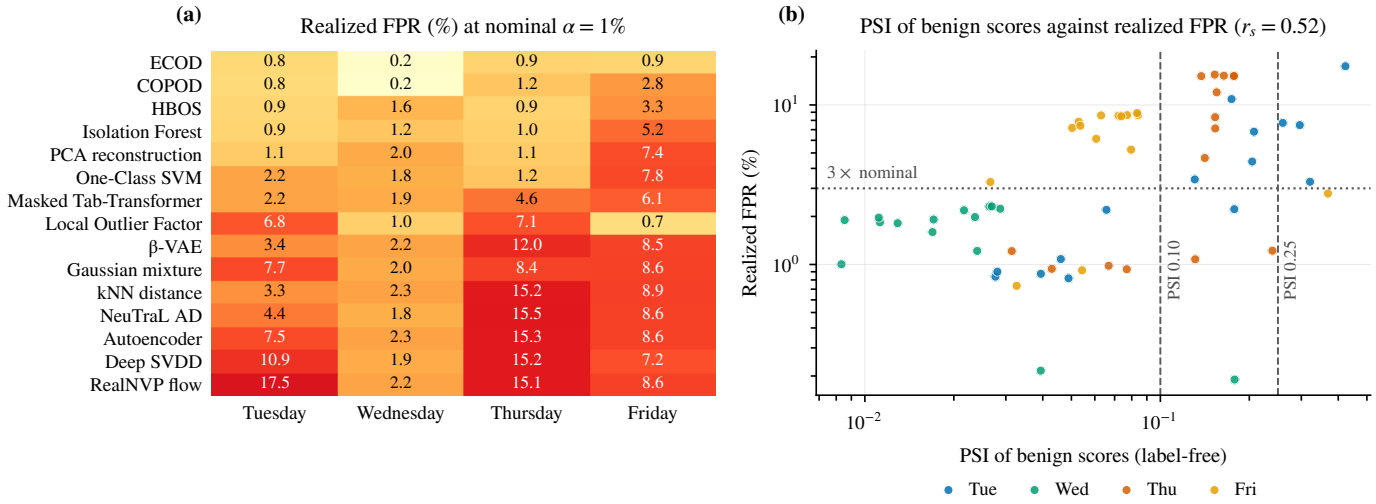


Fig. 5: (a) Realized FPR (%) by day and detector for a Monday-calibrated threshold at nominal $\alpha = 1\%$. (b) PSI of benign scores against realized FPR for every single detector and day; dashed lines mark PSI 0.10 and 0.25, and the dotted line marks $3\times$ the nominal rate.

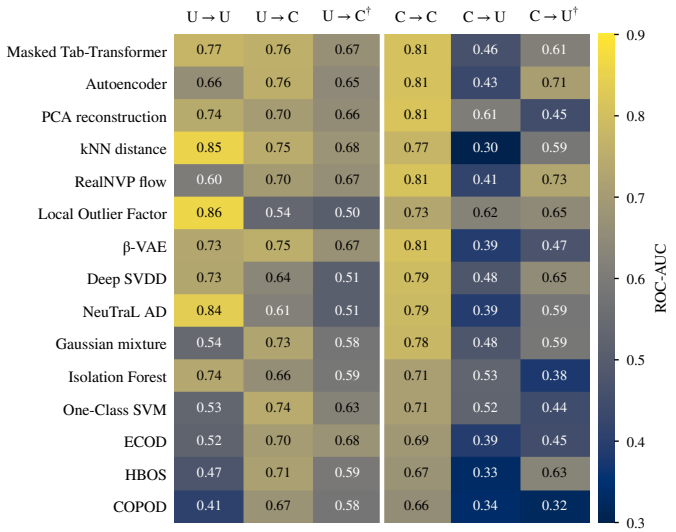


Fig. 6: ROC-AUC for in-dataset and cross-dataset evaluation in the harmonized space (U: UNSW-NB15, C: CIC-IDS2017; † per-domain standardization). Detectors are sorted by mean AUC.

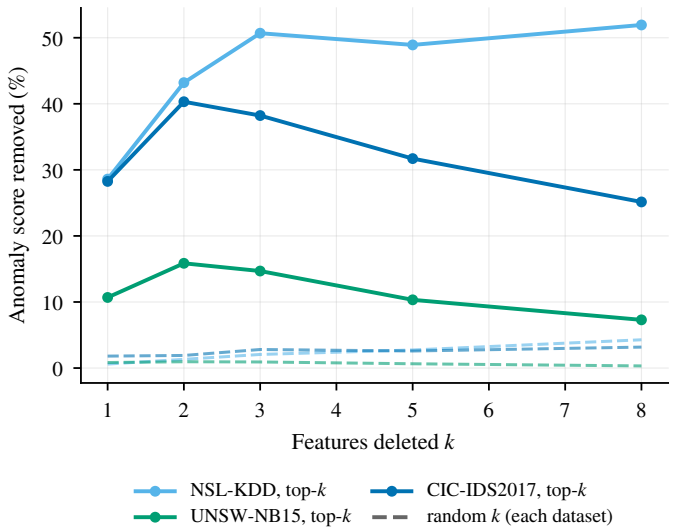


Fig. 7: Deletion faithfulness of autoencoder attributions: share of the anomaly score removed as the top-k attributed features (solid) or k random features (dashed) are deleted.

D. E4: dataset shift (Q3)

Fig. 6: the in-dataset mean AUC on the 18-feature harmonized space is 0.667 for UNSW-NB15 and 0.757 for CIC-IDS2017; transferred, it is 0.695 (UNSW-NB15→CIC-IDS2017) and 0.446 (CIC-IDS2017→UNSW-NB15), the latter below chance. On average, detectors retain 80% of their in-dataset AUC when transferred. The realized FPR at a nominal 1% is 56% and 28% in the two directions, against 2% and 3% in-dataset, so the conformal guarantee collapses entirely. Per-domain standardization (marked †) is not a remedy: it changes the transferred AUC to 0.610 and 0.550, helping one direction and hurting the other.

E. Explaining the autoencoder's alarms

For the autoencoder, the per-feature squared error is an *exact* additive decomposition of the score, so attribution needs no approximation. We test its faithfulness by deletion: setting the top-3 attributed features to the standardized mean (zero) and re-scoring removes, on average, 51%, 15% and 38% of the score on NSL-KDD, UNSW-NB15 and CIC-IDS2017, against 2%, 1% and 3% when three random features are deleted instead (Fig. 7). K-means on the attribution vectors of flagged flows recovers attack categories with normalized mutual information 0.19, 0.41 and 0.46, so attributions carry category-level information that analysts can use for triage.

VII. Discussion

What the evidence supports: Taken together, the experiments say that the dominant risk for an unsupervised detector in the field is not a few points of AUC but the loss of its operating point. A detector that ranks flows well can still flag several times its promised share of benign traffic after a day of drift or on a new network, and the conformal diagnostic makes that failure explicit and measurable. Contamination, by contrast, is partly repairable: CTR recovers roughly half of the average loss at 5% contamination without labels.

Practical guidance: (i) Report the realized false-positive rate on recent held-out benign traffic, not only AUC. (ii) Prefer diverse conformal ensembles over single detectors or over everything at once; the diverse merge was the only configuration that stayed within budget in-dataset while keeping useful recall. (iii) Purify the training window with a committee when attacks cannot be excluded, choosing the trim as an upper bound on plausible contamination. (iv) Monitor the score distribution with PSI and recalibrate when it crosses a threshold tuned on drift cases, rather than the conventional 0.25.

Limitations and threats to validity: All three corpora are lab-generated, have documented issues [10] and do not reflect today's encrypted traffic, so absolute numbers are not production figures; the claims concern *relative* behaviour. Hyper-parameters are fixed across datasets, which is fair but not optimal for any dataset. Seeds are replicates over models and not over datasets, so rank tests are optimistic, and the dataset-level test is not significant (Section VI). A clean calibration slice is assumed. CTR requires an assumed contamination bound and costs accuracy on clean data. The PSI analysis selects benign flows with ground truth to separate drift from prevalence, and the harmonized transfer space is lossy, so a failure to transfer there is evidence about that representation only. "Anomalous" is not "malicious": benign but unusual flows score like attacks, so triage and supervised layers remain necessary. We do not evaluate adaptive adversaries who evade or poison the detectors. TabPFN-style tabular foundation models were examined and excluded because per-feature conditional prediction is impractical on CPU at this scale.

Future work: The natural next steps are conformal methods that adapt to drift [34]–[36] and a drift-triggered recalibration policy, evaluation on modern encrypted-traffic corpora with the protocol unchanged, and cost-sensitive operating points tied to analyst triage budgets.

VIII. Conclusion

Evaluating unsupervised intrusion detectors on a clean training window with one threshold-free number hides exactly the failure modes that matter in deployment. A protocol that injects contamination, replays time and crosses networks, paired with conformal calibration, makes those failures measurable: hidden attacks cost 11.4 AUC points at 5% contamination, a label-free consensus screen recovers roughly half of it, calibrated false-positive rates drift from 1.6% to 7.6% across days, and transfer between networks is poor. A population-stability index of the benign scores anticipates the loss of the guarantee, and diverse conformal ensembles are the most reliable way we found to stay within budget.

Reproducibility and Data Availability

All datasets are public and downloaded with pinned SHA-256 checksums. The code, the analysis of the per-seed score files, the generated tables and figures of this paper, and an interactive laboratory are available at https://github.com/fragompul/unsupervised-anomaly-network-intrusion. A full run (all four experiments, five seeds) takes about three hours on a ten-core CPU laptop; experiment cells are resumable and fault-isolated, and every reported number is recomputed from stored raw scores by a single command.

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